Douglas Miller

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I am a fourth-year undergraduate with research experience in CMP theory and quantum gravity. I intend to pursue a PhD in CMP theory. In particular, I plan to focus my research primarily on topological phases and matter, anyonic physics, and quantum Hall interferometry. My long-term goal is to lead a research group as a theoretical physicist, with a commitment to understanding novel phenomena using creative mathematics while training the next generation of researchers to be even better than ours.

Education

Florida Atlantic University (FAU) Boca Raton, Florida
B.S. in Physics, minor in Mathematics Expected May 2027

- Cumulative GPA: 3.97/4.00
- Senior honors thesis (in progress): *Quantum Gravity: Continuum Reconstruction from Discrete Ponzano–Regge Data*.

Research Experience

Florida Atlantic University, Department of Physics Boca Raton, Florida
Undergraduate Researcher Fall 2025–Present

Advisor: Prof. Jonathan Engle

- Project: Levi-Civita Curvature in Ponzano-Regge.
- Developed code to compute spinfoam amplitudes. Manuscript in progress.

University of Rochester, Department of Physics and Astronomy Rochester, New York
Undergraduate Research Intern | NSF REU and current collaboration Summer 2026–Present

Advisor: Prof. Chaitanya Murthy

- Project: scattering theory for quantum point contacts in quantum Hall Fabry–Pérot interferometry.
- Developed a gauge-independent treatment of electron scattering theory for quantum Hall interferometry.

Talks and Presentations

Talk and Poster presentation | University of Rochester Physics REU Symposium Summer 2026
Gauge-Invariant Treatment of Electron Scattering for Quantum Hall Interferometry

Contributed talk | 12th Gulf Coast Gravity Meeting, Troy, Alabama April 17–18, 2026
Expectation Value of Levi-Civita Curvature in Ponzano-Regge: Non-flatness in 3D Quantum Gravity

Research Collquium | Department of Physics, FAU April 2026
Expectation Value of Levi-Civita Curvature in Ponzano-Regge

Dr. Eric Shaw Excellence in Science Undergraduate Research Poster Competition | FAU October 2025
Curvature in Ponzano–Regge 3D Quantum Gravity

Honors and Awards

Bjorn Lamborn Scholarship in Physics Spring 2026
- Excellence in Theoretical Physics

Florida Atlantic University Honors Program 2023–2024
- Completed the two-year program in one year.

President's List Fall 2023, Spring 2025, Fall 2025, Spring 2026
- Awarded for attaining a semester GPA of 4.00.

Dean's List Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026
- Awarded for attaining a semester GPA of at least 3.75.

Teaching and Outreach

Physics and Mathematics Tutor August 2024–Present
- Tutor high-school and college students in calculus and introductory physics courses. Pre-calculus, Calculus 1, College physics 1,2.

Annual Pumpkin Drop and Physics Carnival, FAU Department of Physics October 2023–Present
- Assist with event preparation and outreach activities. Help host 400 4th, 5th, and 6th graders.

Loops '24 International School and Conference on Loop Quantum Gravity Spring 2024
- Provided volunteer and logistics support for the international school and conference.

Technical Skills and Languages

Computer programming: Python (NumPy, SciPy, Matplotlib, SymPy), Mathematica, HTML, CSS

Typesetting: LaTeX

Languages: English (fluent), Polish (fluent), Spanish (elementary), German (elementary)

Relevant Course-Work

Graduate Level: Mathematical Physics 1 (functional analysis) | Quantum Mechanics | Algebraic Topology | Classical Mechanics

Undergraduate Level: General Relativity (directed independent study) | Complex Analysis | Discrete Mathematics